Fiscal Stimulus in a Monetary Union

Evidence from US Regions

Nakamura and Steinsson (2014), *American Economic Review*

Topics in Macroeconomics — Week 2 reading group

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How large is the government spending multiplier?

- The ARRA (2009) was motivated by a multiplier of 1.6 [Romer & Bernstein (2009)]; others argue it is close to zero
- **Hard to measure:** government spending is rarely exogenous
  - Wars [Ramey & Shapiro (1998); Ramey (2011); Barro & Redlick (2011)]: few, and confounded by taxes, price controls, patriotism, and other macroeconomic shocks
  - Structural VARs [Blanchard & Perotti (2002); Mountford & Uhlig (2009)]: rely on structural assumptions about output and fiscal dynamics
- **Hard to interpret:** the multiplier is not a deep structural parameter
  - Small in the Neoclassical model [Baxter & King (1993), < 0.5]
  - In New Keynesian (NK) models it depends on how much monetary policy "leans against the wind": the more counter-cyclical the policy, the smaller the multiplier [Eggertsson (2010); Christiano, Eichenbaum & Rebelo (2011)]
- **Question:** what does government spending in one region do to its output relative to other regions, when relative monetary and tax policy cannot respond?

$\Rightarrow$ US is a monetary and fiscal union: relative policy is pinned down

The empirical evidence, and how they get it

- **Data:** military procurement by state, 1966–2006; state GDP, employment, prices, population
- **Variation:** national military buildups and drawdowns move spending far more in some states
  - National spending +1pp of GDP: California about +3pp of its GDP, Illinois less than +0.5pp
- **Method:** two-stage least squares on a panel of states (and regions), with state and year fixed effects
- **Findings:**
  1. Open economy relative multiplier ≈ 1.5 for output; 1.3–1.8 for employment
     - When relative per capita government purchases in a region rise by 1% of regional output, relative output per capita in that region rises by about 1.5%
  2. Relative prices and population barely move over two years
  3. Output multiplier larger when unemployment is high

The model, and what each ingredient does

- Two-region NK model of a monetary and fiscal union, used to interpret the estimate
- **Time fixed effects:** the relative multiplier "differences out" aggregate monetary and tax policy
  - 0.83 under every monetary rule, while the closed economy aggregate multiplier ranges from 0.20 to ∞
- **Fixed exchange rate within the union:** relative prices must come back down
  - The long-term relative real rate does not fall: a monetary union is not a zero lower bound
- **Fundamentals decide the size:**
  - Separable utility: 0.83 (relative consumption falls); Neoclassical: below 0.5
  - Consumption and labor as complements (GHH): 1.42, close to the estimate

$\Rightarrow$ The evidence favors models in which demand shocks can have large effects on output

Contribution to the literature

- **Empirical fiscal multipliers** [Ramey (2011); Barro & Redlick (2011); Blanchard & Perotti (2002)]
  - ⇒ New: regional variation is far larger than aggregate variation
- **Multipliers and policy** [Woodford (2011); Eggertsson (2010); Christiano, Eichenbaum & Rebelo (2011)]
  - ⇒ New: an estimand invariant to aggregate monetary and tax policy
- **Regional multipliers and monetary unions** [Chodorow-Reich et al. (2012); Shoag (2010); Benigno & Benigno (2003); Galí & Monacelli (2008); Farhi & Werning (2012)]
  - ⇒ New: a framework mapping relative into aggregate multipliers; it equals the multiplier of a small open economy with a fixed exchange rate [cf. Ilzetzki, Mendoza & Végh (2013): 1.5 under pegs]

What I like and what I don't like

Like

- **New data:** Census shipments digitized from microfilm check subcontracting ($\hat{\beta} = 0.96$); state procurement 1966–2006 extends earlier work [Davis, Loungani & Mahidhara (1997)]
- **Identification:** Variation used is easy to gauge and explicit in a simple graph; the key assumption is weak
- **Model fundamentals:** Search across various model fundamentals (sticky vs. flexible prices, monetary rules, fiscal rules, preferences)

Don't like

- **Few effective shocks:** variation used dominated by "few buildups and drawdowns"; exact number not provided
- **Weak instruments:** 51 instruments, Cragg–Donald $F \approx 5$ [Stock & Yogo (2005)]
- **GHH vs. HTM:** GHH matches 1.5, but hand-to-mouth households may too [Galí, López-Salido & Vallés (2007)]: would they imply the same aggregate multiplier?

Roadmap

1. **Empirical results**
   - Data, identification, instruments, regression, main result, robustness
2. **Model**
   - Environment, households, preferences, government, firms, equilibrium, calibration
3. **Theoretical results**
   - Mechanism, aggregate vs relative multiplier, GHH preferences, extensions

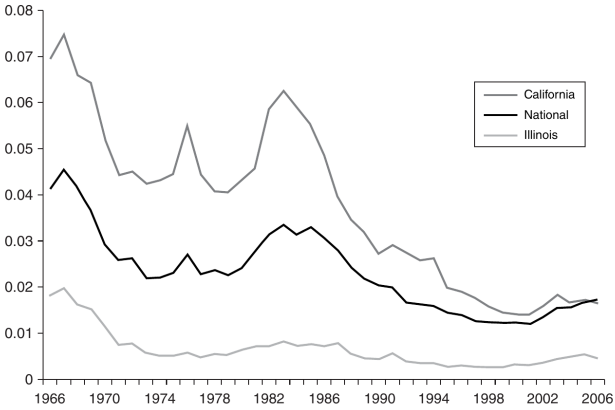
Empirical results

Data

Series	Source	Coverage
Military procurement	DD-350 prime contracts (Dept. of Defense) 90% of purchases; >$10k (>$25k after 1983)	1966–2006, fiscal year
Military shipments	Census, defense-oriented industries digitized from microfilm	1963–1983
Output	BEA GDP by state, and by sector	since 1963
Employment	BLS payroll survey (CES); BEA as a check	since 1969 (BEA)
Prices	State CPI: Del Negro (1998) to 1995, ACCRA after	1969–2006
Population	Census Bureau	annual

- Units: 51 states, and 10 regions (census divisions, with the South Atlantic split in two)
- All variables per capita; output and spending deflated by the national CPI
- DD-350 records where the prime contractor does most of the work; the military shipments data assesses subcontracting; they track each other ($\hat{\beta} = 0.96$) Appendix

Fact: some states are far more exposed to national buildups



[Figure 1: prime military contracts / state GDP]

- National spending is driven by geopolitics: Vietnam, the Soviet invasion of Afghanistan, 9/11
- Vietnam drawdown, 1966–71: California −2.5pp, almost twice the national fall; Illinois about −1pp
- Most sensitive states: Missouri, Connecticut, Texas, Vermont, New Hampshire, Massachusetts, Kansas, California, Georgia, Louisiana

[Table 1]

Identification: differential exposure to national buildups

- **Identifying assumption:** the US does not launch military buildups because the states that receive a disproportionate share of military spending are doing badly relative to other states

- Similar but weaker than the usual assumption that national military spending is exogenous to the US business cycle

- Use this variation to instrument government spending with differential sensitivity of military spending across regions to national military buildups and drawdowns

Shift-share design. One national shift per year (procurement) times state exposure — estimated in the first stage (baseline) or the 1966–71 spending share (Bartik)

[Section II.A]

Two instruments

- **Baseline:** national procurement × state dummy
  - The first stage estimates each state's sensitivity to national buildups
- **Bartik-type:** national procurement × the state's 1966–71 spending relative to its output [Bartik (1991)]

	Baseline		Bartik-type	
	States	Regions	States	Regions
Partial R^2 of excluded instruments	12%	18%		
Cragg–Donald first-stage F	≈ 5	≈ 8	106	53

- Monte Carlo: the baseline state regression is biased toward OLS, by about 10% (true ≈ 1.65)
- Fixed effects IV with multipliers that differ across states [Murtazashvili & Wooldridge (2008)]

[Sections II.A, II.C; footnotes 23, 24, 30]

The regression

$$\frac{Y_{it} - Y_{it-2}}{Y_{it-2}} = \alpha_i + \gamma_t + \beta \frac{G_{it} - G_{it-2}}{Y_{it-2}} + \varepsilon_{it} \qquad (1)$$

- y: two-year change in per capita output, relative to its initial level
- x: two-year change in per capita military procurement, divided by initial output
- z: national procurement $\times$ state dummy (baseline), or $\times$ initial spending share (Bartik)
- Controls: state fixed effects α_i (allow state-specific trends) and year fixed effects γ_t (absorb aggregate shocks and aggregate policy: the Fed, federal taxes)
- Horizon: two years — a parsimonious way to capture the dynamics; overlapping observations, so standard errors are clustered by state (region)
- Sample: 1966–2006; 51 states ($N = 1{,}989$) or 10 regions ($N = 390$)

$\Rightarrow$ β is the open economy relative multiplier

Main result: open economy relative multiplier ≈ 1.5

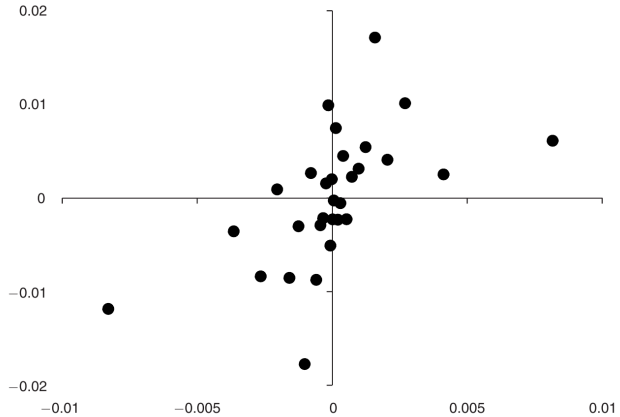
	Output		Output, defl. state CPI		Employment		CPI	Population
	States	Regions	States	Regions	States	Regions	States	States
Prime military contracts	1.43	1.85	1.34	1.85	1.28	1.76	0.03	−0.12
	(0.36)	(0.58)	(0.36)	(0.71)	(0.29)	(0.62)	(0.18)	(0.17)
Prime contracts plus military compensation	1.62	1.62	1.36	1.44	1.39	1.51	0.19	0.07
	(0.40)	(0.84)	(0.39)	(0.96)	(0.32)	(0.91)	(0.16)	(0.21)
Observations	1,989	390	1,989	390	1,989	390	1,763	1,989

2SLS; state (region) and year fixed effects; standard errors clustered by state (region); 1966–2006 (CPI: 1969–2006).

- Relative spending +1% of output ⇒ relative output +1.4% (states) to +1.9% (regions); employment +1.3% to +1.8%
- Relative prices and population: no significant response over two years
- Output multiplier larger when unemployment is high (3.5–4.3; $p \approx 0.06$–0.07), not employment; largest in construction and manufacturing Appendix

[Table 2]

The IV in one picture



[Figure 3]

- Vertical: two-year change in output; horizontal: predicted change in military spending (first stage)
- 30 quantile bins; both demeaned by year and state fixed effects
- Most points sit in the north-east and south-west quadrants $\Rightarrow \beta > 0$
- Outliers: Missouri and Connecticut (most sensitive), North Dakota (negative sensitivity); dropping any of them raises β — all three: 1.88 (0.57)

[Footnote 28]

Robustness

- **Bartik-type instrument:** output 2.48 (0.94) states, 2.75 (0.69) regions; employment 1.81, 2.51
- **Specifications:** output per working-age person 1.46; oil price × state controls 1.32; real rate × state controls 1.40; BEA employment 1.52
- **LIML**, robust to many instruments: 1.95 (0.62)
- **OLS:** 0.16 (0.14), far below IV — endogenous allocation and attenuation bias; with shipments as the instrument (1966–82): IV 1.3 vs OLS 0.2
- **Cyclicality:** high- and low-military states have the same volatility of output growth
- **CPS employment:** 1.4 (0.5) vs CES 1.8 (0.4) on the same sample
- **Anticipation:** adding future spending raises the summed effect [Ramey (2011)]

[Table 3; Section II.C]

Model

Environment

- **Two regions in a monetary and fiscal union** — home (population n) and foreign $(1-n)$
  - In each region, a continuum of household types x, each supplying a differentiated labor input
  - A continuum of firms z, each producing one variety
  - A federal government: buys goods in both regions, sets taxes, runs monetary policy
  - Household preferences, market structure, and firm behavior take the same form in both regions
- **Decisions:**
  - Households: consumption with home bias, hours, state-contingent assets (complete markets across regions in the baseline)
  - Firms: labor demand and Calvo price setting
  - Government: AR(1) spending in each region; lump-sum or balanced-budget taxes; a Taylor rule
- Goods trade freely across regions; labor is immobile; capital is fixed in the baseline

Households

Household of type x maximizes $E_0 \sum_{t=0}^{\infty} \beta^t u(C_t, L_t(x))$ (2) over the consumption index (3), subject to the budget constraint (4):

$$C_t = \left[\phi_H^{\frac{1}{\eta}} C_{Ht}^{\frac{\eta-1}{\eta}} + \phi_F^{\frac{1}{\eta}} C_{Ft}^{\frac{\eta-1}{\eta}} \right]^{\frac{\eta}{\eta-1}} \tag{3}$$

$$P_t C_t + E_t[M_{t,t+1} B_{t+1}(x)] \leq B_t(x) + (1-\tau_t) W_t(x) L_t(x) + \int_0^1 \Xi_{ht}(z)\, dz - T_t \tag{4}$$

Optimality: Euler equation, labor supply, demands FOCs. Combining home and foreign Euler equations:

$$\frac{u_c(C_t^*, L_t^*(x))}{u_c(C_t, L_t(x))} = Q_t, \qquad Q_t = \frac{P_t^*}{P_t} \tag{5}$$

- $\phi_H > n$: home bias; η: elasticity between home and foreign goods
- (5) is the Backus–Smith risk-sharing condition [Backus & Smith (1993)]: with separable preferences, if home goods become more expensive, home consumption falls relative to foreign

[Section III.A]

Preferences: separable vs GHH

Separable, and GHH [Greenwood, Hercowitz & Huffman (1988)] where consumption and labor are complements:

$$u(C_t, L_t(x)) = \frac{C_t^{1-\sigma^{-1}}}{1-\sigma^{-1}} - \chi \frac{L_t(x)^{1+\nu^{-1}}}{1+\nu^{-1}} \tag{6}$$

$$u(C_t, L_t(x)) = \frac{\left(C_t - \chi L_t(x)^{1+\nu^{-1}}/(1+\nu^{-1})\right)^{1-\sigma^{-1}}}{1-\sigma^{-1}} \tag{7}$$

$\Rightarrow$ Separable: relative consumption must fall when home goods get more expensive; GHH: it can rise

Government

Government: Consumes goods and conducts fiscal and monetary policy

Spending: G_{Ht}, G_{Ft} follow AR(1) processes, with the same CES demand as households

Fiscal policy: (i) Government spending shocks fully financed by lump-sum taxes; (ii) "balanced budget" tax policy where labor income taxes vary

Monetary policy:

$$\hat{r}_t^n = \rho_r \hat{r}_{t-1}^n + (1-\rho_r)\left(\phi_\pi \hat{\pi}_t^{ag} + \phi_y \hat{y}_t^{ag} + \phi_g \hat{g}_t^{ag}\right) \qquad (8)$$

- Three rules, successively less "leaning against the wind":
  - Volcker–Greenspan — $\rho_r = 0.8$, $\phi_\pi = 1.5$, $\phi_y = 0.5$, $\phi_g = 0$ [Clarida, Galí & Gertler (2000)]
  - Fixed real rate — fixed real interest rate in response to government spending shocks; aggressive response to inflationary effects of other shocks
  - Fixed nominal rate — fixed nominal interest rate in response to government spending shocks; aggressive response to inflationary effects of other shocks (proxy for the ZLB)

[Sections III.B, IV]

Firms

Continuum indexed by z in the home region, where firm z specializes in the production of differentiated good z

Production: $y_{ht}(z) = f(L_t(z)) = L_t(z)^a$ using a fixed, nondepreciating stock of capital and labor

Competition: Each firm belongs to an industry x and there are many firms in each industry; goods in industry x are produced with labor of type x, and all firms in industry x change prices at the same time [Woodford (2003)]

[Sections III.C–D]

Firms: decision

Firm z in industry x maximizes $E_t \sum_{j=0}^{\infty} M_{t,t+j}[p_{ht+j}(z)y_{ht+j}(z) - W_{t+j}(x)L_{t+j}(z)]$ (9) subject to (10) as the firm must satisfy demand from home consumers, foreign consumers, and the government:

$$y_{ht}(z) = \left(nC_{Ht} + (1-n)C^*_{Ht} + nG_{Ht}\right)\left(\frac{p_{ht}(z)}{P_{Ht}}\right)^{-\theta} \leq f(L_t(z)) \quad (10)$$

Optimal labor choice: $W_t(x) = f_l(L_t(z))S_t(z)$ (11), where $S_t(z)$ denotes the firm's nominal marginal cost

Calvo pricing: reoptimize price with probability $1-\alpha$; optimal price setting follows a markup over expected marginal cost:

$$p_{ht}(z) = \frac{\theta}{\theta - 1} E_t \sum_{j=0}^{\infty} \frac{\alpha^j M_{t,t+j} y_{ht+j}(z)}{E_t \sum_{k=0}^{\infty} \alpha^k M_{t,t+k} y_{ht+k}(z)} S_{t+j}(z) \quad (12)$$

[Sections III.C–D]

Equilibrium

Definition (our statement of Section III). Given $\{G_{Ht}, G_{Ft}\}$ and the policy rules, an equilibrium is a sequence of

- allocations $\{C_t, C_{Ht}, C_{Ft}, L_t(x), y_{ht}(z), B_{t+1}(x)\}$,
- prices $\{p_{ht}(z), P_{Ht}, P_t, W_t(x), \hat{r}_t^n, Q_t\}$,
- taxes $\{\tau_t, T_t\}$,

for home and foreign, such that

1. households maximize (2) subject to (4), given prices;
2. firms choose labor (11) and reset prices (12) under Calvo, subject to demand (10);
3. the central bank follows (8); taxes follow the lump-sum or balanced-budget rule;
4. goods markets clear ((10) with equality), labor markets clear for each type x, and asset markets clear.

[Sections III, IV]

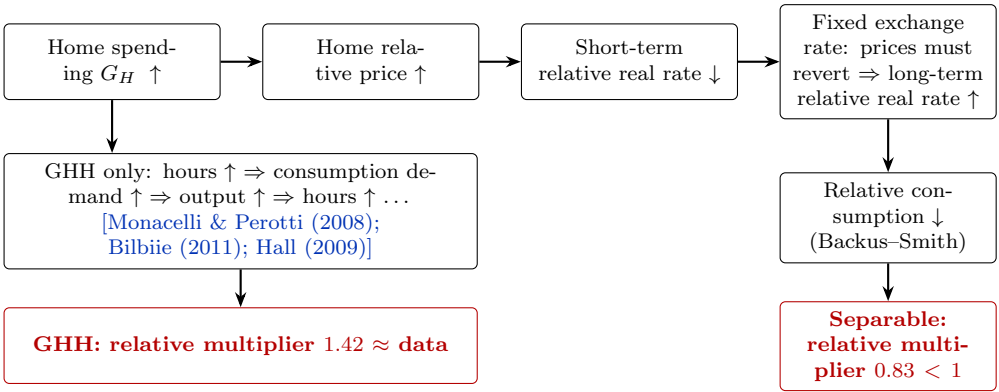
Calibration

Parameter (quarterly)	Meaning	Target or source
$\beta = 0.99$	discount factor	
$\sigma = 1$	IES	balanced growth with separable preferences
$\nu = 1$	Frisch elasticity	extensive margin [Hall (2009); Chetty et al. (2011)]
$\theta = 7$	across varieties	
$\eta = 2$	home vs foreign goods	as in [Obstfeld & Rogoff (2005)]
$a = 2/3$	$f(L) = L^a$	
$\alpha \in \{0, 0.75\}$	Calvo: flexible, sticky	sticky: prices reset once a year on average
$n = 0.1$	home region size	average region in the regressions
$G/Y = 0.2$	steady state	
$\phi_H = 0.69$	home bias	inter-region trade 31% of GDP (CFS, NIPA)
$\rho_g = 0.933$	persistence of G	procurement data, simulated method of moments

[Sections III.D, IV; online Appendix B]

Theoretical results

The mechanism



[Section IV.B, Figures 4–5; the same logic holds in a small open economy with a fixed exchange rate [Corsetti, Kuester & Müller (2011)]]

Aggregate vs relative multiplier: separable preferences

	Aggregate (closed economy)	Relative (open economy)
Sticky prices		
Volcker–Greenspan	0.20	0.83
Fixed real rate	1.00	0.83
Fixed nominal rate	∞	0.83
with $\rho_g = 0.85$	1.70	0.90
Flexible prices		
Constant tax rates	0.39	0.43
Balanced budget	0.32	0.43

[Table 6]

- **Aggregate:** from 0.20 to ∞, with the monetary rule
  - $= 1$ under a fixed real rate [Woodford (2011)]
  - Very large at a fixed nominal rate [Eggertsson (2010); Christiano, Eichenbaum & Rebelo (2011)]
  - Lower with distortionary taxes [Baxter & King (1993)]
- **Relative:** 0.83 under every monetary rule; 0.43 under either tax rule
- Neither relative multiplier is close to the estimate: 1.34 (states) to 1.85 (regions) with output deflated by the state CPI, as in the model

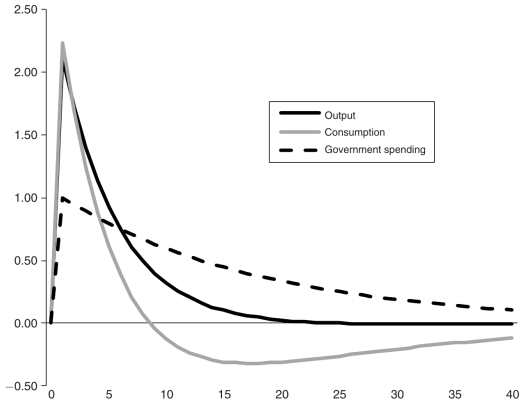
GHH preferences match the data

	Aggregate (closed economy)	Relative (open economy)
Sticky prices		
Volcker–Greenspan	0.12	1.42
Fixed real rate	7.00	1.42
Fixed nominal rate	∞	1.42
with $\rho_g = 0.50$	8.73	2.04
Flexible prices		
Constant tax rates	0.00	0.30
Balanced budget	-0.18	0.30

[Table 7]

- **Relative:** 1.42 under every monetary rule, close to the estimate of 1.5; 2.04 with more transitory shocks ($\rho_g = 0.50$)
- **Neoclassical** with GHH: 0.30
- **Aggregate** under Volcker–Greenspan: only 0.12 — GHH does not raise the aggregate multiplier in general

GHH: relative output and consumption after a spending shock



- Output and consumption rise on impact by a little more than twice the shock, then fall faster than the shock
- Consumption rises as much as output: home runs a trade deficit, then a surplus once consumption falls below steady state
- Complementarity: households shift consumption toward periods of high work effort

[Figure 6]

Extensions: neither changes the conclusion

- **Incomplete markets** (separable): only a noncontingent bond is traded across regions; spending financed locally or federally [Farhi & Werning (2012)]
  - Federally financed: $0.83 \to 0.90$; reaching 1.5 would need $\eta < 0.6$ Appendix
- **Variable capital** (GHH)
  - Firm-specific capital [Woodford (2003, 2005)]: 1.47, close to the baseline 1.42
  - A frictionless regional rental market [Christiano, Eichenbaum & Evans (2005)]: 0.98; firm-specific capital helps New Keynesian models match sluggish prices [Eichenbaum & Fisher (2007); Altig et al. (2011)] Appendix

$\Rightarrow$ Separable preferences stay far below 1.5; GHH with firm-specific capital stays close to it

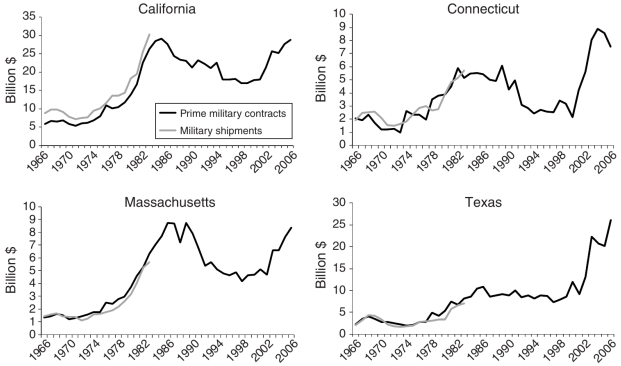
[Sections IV.D–IV.E; Tables 8–9]

Summary of the paper

- $\Rightarrow$ When relative military spending rises by 1% of output, relative output rises by about 1.5%; employment similarly; prices and population barely move
- $\Rightarrow$ The open economy relative multiplier differences out aggregate monetary and tax policy: a diagnostic tool, not a closed economy multiplier
- $\Rightarrow$ Separable New Keynesian model: 0.83; Neoclassical: below 0.5; New Keynesian with consumption–labor complementarity: 1.42
- $\Rightarrow$ Demand shocks can have large effects on output; how large the aggregate multiplier is depends on monetary policy

Appendix

Does procurement measure local production?



[Figure 2]

- Prime contracts record the main contractor, not its subcontractors in other states
- Check: Census shipments to the government from defense-oriented industries, 1963–83
- Regress shipments on procurement: $\hat{\beta} = 0.96$
- The remaining gap is measurement error in both series — which the instruments also correct

Back

Results by sector and by unemployment state

By sector (states)

	Weight	Multiplier	
Construction	0.05	5.43*	(1.24)
Manufacturing	0.20	2.83*	(0.95)
Retail	0.09	1.36*	(0.28)
Services	0.18	0.99*	(0.39)
Government	0.13	0.15	(0.34)
Federal military	0.01	0.23	(0.82)

- Government output adds almost nothing; no rise in other military output

[Table 4; * = significant at 5%]

By slack (states: β_h, β_l high and low unemployment)

	Output		Employment	
	National	State	National	State
β_h	3.54	4.31	1.85	1.32
	(1.55)	(1.80)	(0.87)	(0.81)
$\beta_l - \beta_h$	−2.80	−3.37	−0.75	0.03
	(1.49)	(1.84)	(0.89)	(0.84)

- Output: 3.5–4.3 in high slack vs 0.7–0.9 in low slack; difference $p \approx 0.06$–0.07
- In line with larger multipliers in recessions [Auerbach & Gorodnichenko (2012); Shoag (2010)]

[Table 5]

Back

Households: optimality conditions

Euler equation, state by state for all $j > 0$, with a standard transversality condition:

$$\frac{u_c(C_{t+j}, L_{t+j}(x))}{u_c(C_t, L_t(x))} = \frac{M_{t,t+j}}{\beta^j} \frac{P_{t+j}}{P_t} \qquad (A1)$$

Labor supply, the intratemporal trade-off between consumption and hours:

$$\frac{u_\ell(C_t, L_t(x))}{u_c(C_t, L_t(x))} = (1 - \tau_t) \frac{W_t(x)}{P_t} \qquad (A2)$$

Demands for the home and foreign composites, minimizing the cost of attaining C_t:

$$C_{Ht} = \phi_H C_t \Big(\frac{P_{Ht}}{P_t}\Big)^{-\eta}, \qquad C_{Ft} = \phi_F C_t \Big(\frac{P_{Ft}}{P_t}\Big)^{-\eta} \qquad (A3)$$

- Under GHH (7), the marginal rate of substitution between hours and consumption depends on hours only: no wealth effect on labor supply [Greenwood, Hercowitz & Huffman (1988)]

[Section III.A, eqs. (8)–(10)]

Extension: incomplete markets

Relative multiplier, separable preferences	Sticky prices	Flexible prices
Complete markets (baseline)	0.83	0.43
Incomplete markets, locally financed	0.84	0.41
Incomplete markets, federally financed	0.90	0.40

[Table 8]

- With incomplete markets, how spending is financed can matter in a monetary union [Farhi & Werning (2012)]; here only a noncontingent bond is traded across regions
- **Federally financed:** home households are wealthier; with home bias, home demand rises — but only to 0.90
- **Locally financed:** still slightly above the baseline, since under complete markets a home spending shock implies a negative transfer to home
- Reaching 1.5 would need $\eta < 0.6$, below most estimates ($\eta = 1$ gives 1.05)

[Section IV.D]

Back

Extension: variable capital

Relative multiplier, GHH preferences	Output	CPI inflation
Baseline (fixed capital)	1.42	0.17
Firm-specific capital	1.47	0.15
Regional capital market	0.98	0.09
Firm-specific capital, flexible prices	0.25	0.36

[Table 9]

- **Firm-specific capital** with convex adjustment costs [Woodford (2003, 2005)]: close to the baseline; firms invest when the shock hits
- **Regional rental market** [Christiano, Eichenbaum & Evans (2005)]: firms that raise their price can rent less capital at no cost, which weakens strategic complementarity in pricing [Eichenbaum & Fisher (2007); Altig et al. (2011)]
- **Inflation:** sticky prices give a small rise in relative inflation, as in the data; flexible prices a much sharper one

[Section IV.E]

Back

What the relative multiplier says about the aggregate multiplier

- It equals the aggregate multiplier of a small open economy with a fixed exchange rate
  - Across 44 countries: about 1.5 under pegs, much lower under floats [Ilzetzki, Mendoza & Végh (2013)]
- Relative monetary policy in a union is more accommodative than Volcker–Greenspan
  - 1.5 is consistent with low aggregate estimates under normal policy [Barro & Redlick (2011)]
- The model that fits (New Keynesian with GHH) implies large aggregate multipliers when monetary policy does not respond, as at the zero lower bound
- **Welfare** [Woodford (2011)]:

$$\frac{dU}{dG} = \left(U_C - \frac{-U_L}{f_L}\right)\frac{dY}{dG} + (U_G - U_C) \qquad (A4)$$

  - The multiplier matters only through the first term, which is non-zero when output is inefficient
  - "Opportunistic" vs "stimulus" spending at the zero lower bound [Werning (2012)]

[Introduction; Section IV.F]